

The empirical recurrence for OEIS A282860: recovering a conjecture that is not written down

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Abstract

OEIS A282860 records “Empirical recurrence of order 85”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 103$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 85, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 85 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282860 is “Number of $n \times 6$ 0..1 arrays with no 1 equal to more than one of its horizontal, vertical and antidiagonal neighbors.”. It has offset 1 and begins

44, 715, 11250, 197892, 3303160, 55930891, 945547378, 15974465522, ...

The entry states:

Empirical recurrence of order 85 (see link above).

As of the “Last modified” line on the live entry (Jun 09 2023 22:02:03, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 759$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 103$ of the 759, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 103$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 206$ exact terms of the model returns order 85 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{85} c_i a(n-i),$$

c_1	1	c_{23}	1566624392554	c_{45}	4048637565303	c_{67}	431666301
c_2	148	c_{24}	−994471986653	c_{46}	6115998510272	c_{68}	−30925623
c_3	1448	c_{25}	11883254178	c_{47}	−10261822641994	c_{69}	−40240674
c_4	7792	c_{26}	915956554836	c_{48}	2496727842639	c_{70}	32303615
c_5	31431	c_{27}	−1161114120336	c_{49}	6712952674242	c_{71}	347792
c_6	92281	c_{28}	522278047071	c_{50}	−6528878850185	c_{72}	−1828283
c_7	152115	c_{29}	317799443889	c_{51}	101781144254	c_{73}	1365842
c_8	−33587	c_{30}	−591220562420	c_{52}	3644055833489	c_{74}	118543
c_9	−708081	c_{31}	400201305326	c_{53}	−2191855608659	c_{75}	−111089
c_{10}	−3344840	c_{32}	46626572880	c_{54}	−442948252270	c_{76}	33022
c_{11}	241251	c_{33}	−731594219653	c_{55}	1143333428167	c_{77}	5328
c_{12}	6396147	c_{34}	491379198487	c_{56}	−437431359194	c_{78}	−3323
c_{13}	12666590	c_{35}	1530511163875	c_{57}	−207162069411	c_{79}	1163
c_{14}	57526071	c_{36}	−2597668986983	c_{58}	234935763227	c_{80}	245
c_{15}	374364316	c_{37}	−211580357318	c_{59}	−49404030702	c_{81}	63
c_{16}	−6719104537	c_{38}	3634583703962	c_{60}	−43756598450	c_{82}	37
c_{17}	38499642796	c_{39}	−2290687273250	c_{61}	35798284826	c_{83}	13
c_{18}	−140236702969	c_{40}	−1446555857321	c_{62}	−4382166859	c_{84}	1
c_{19}	373563208381	c_{41}	1103785252645	c_{63}	−6305806281	c_{85}	1
c_{20}	−767507547824	c_{42}	1320587945105	c_{64}	4298366647		
c_{21}	1246341630696	c_{43}	1376962028381	c_{65}	−485376211		
c_{22}	−1600881319677	c_{44}	−6711464944437	c_{66}	−655985246		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 85, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $85 + 4$ terms removed returns order 85 as well, so $\deg q_{\min} = 85 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $85 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 85 that this sequence satisfies — and that family turns out to have exactly one member.

References

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